ROSSMANN STORE SALES
DATA CLEANING REPORT

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1. DATASET OVERVIEW

Dataset	Rows	Columns	Duplicate Rows	Missing Cells
Train	1017209	9	0	0
Test	41088	8	0	11
Store	1115	10	0	2343
Sample Submission	41088	2	0	0

2. MISSING VALUES ANALYSIS

Dataset	Column	Missing Count	Missing Percentage (%)
Test	Open	11	0.03
Store	CompetitionDistance	3	0.27
Store	CompetitionOpenSinceMonth	354	31.75
Store	CompetitionOpenSinceYear	354	31.75
Store	Promo2SinceWeek	544	48.79
Store	Promo2SinceYear	544	48.79
Store	PromoInterval	544	48.79

3. DUPLICATE ANALYSIS

Train: 0
Test: 0
Store: 0

4. MISSING VALUE HANDLING

- CompetitionDistance: Median imputation.
- Missingness indicator created.
- Promo2 missing values handled as structurally not applicable.
- Competition opening information handled using indicators.
- Missing Test Open values inferred using Store and DayOfWeek history.

5. OUTLIER ANALYSIS

Column	Q1	Q3	IQR	Lower Bound	Upper Bound	Outlier Count	Outlier Percentage
sales	4859.00	8360.00	3501.00	-392.50	13611.50	30769	3.64
customers	519.00	893.00	374.00	-42.00	1454.00	40853	4.84

Outliers were detected using the IQR method.

Genuine business outliers were retained.
No automatic deletion or capping was performed.

6. BEFORE AND AFTER COMPARISON

Dataset	Rows Before	Rows After	Columns Before	Columns After	Missing Before	Missing After	Exact Duplicates Before	Exact Duplicates After
Train	1017209	1017209	9	9	0	0	0	0
Test	41088	41088	8	8	11	0	0	0

Store	1115	1115	10	13	2343	0	0
0							

7. FINAL DATA QUALITY VALIDATION

	Count
Train missing cells	0
Test missing cells	0
Store missing cells	0
Train exact duplicates	0
Test exact duplicates	0
Store exact duplicates	0
Invalid train dates	0
Invalid test dates	0
Test Open missing	0
Negative Sales	0
Negative Customers	0
Invalid train DayOfWeek	0
Invalid test DayOfWeek	0

8. CLEANING QUALITY SUMMARY

Dataset	Rows	Columns	Missing Values	Duplicate Rows
Train	1017209	9	0	0
Test	41088	8	0	0
Store	1115	13	0	0

9. EXPORTED CLEANED DATASETS

- cleaned_train.csv
- cleaned_test.csv
- cleaned_store.csv

10. CONCLUSION

The Rossmann datasets were cleaned and validated.

Missing values, duplicates, data types, business rules and outliers were analysed.

The cleaned datasets are ready for further Exploratory Data Analysis and Sales Prediction.

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END OF REPORT